

CHAO HOU, Ph.D.

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RESEARCH INTERESTS

AI-based biomolecule modeling from first principles: sequence → structure ensemble → function / fitness → evolution

AI-based whole-cell modeling linking molecular mechanisms to cellular phenotypes

Phase separation of intrinsically disordered regions

Predicting the effects of genetic variants for precision medicine

Representation and generative deep learning of biological data

EDUCATION AND WORKING EXPERIENCE

Postdoctoral Scientist, Columbia University Sep 2023 – Present

Advisor: Yufeng Shen

Ph.D. of Biomedical Informatics, Peking University Sep 2020 – Jul 2023

Advisor: Tingting Li

Thesis: Identifying phase-separating proteins and characterizing their degradation regulation with machine learning.

Bachelor of Medicine and Economics (double major), Peking University Sep 2015 – Jul 2020

Clinical Internship, Peking University Third Hospital Aug 2018 – Jan 2019

PUBLICATIONS

(# co-first author; * corresponding/co-corresponding author)

Selected:

- Rare inherited variants explain a third of attributable risk for autism.*
Qiao L#, Feliciano P#, Hagen J, **Hou C**, ..., Shen Y*, Chung W*. under revision at *Nature*.
- MotifAE Reveals Functional Sequence Patterns from Protein Language Model: Unsupervised Discovery and Interpretability Analysis.*
Hou C*, Di L, Shen Y*. *Nature Communications* (accepted). [BioRxiv](#), Nov 2025.
- Understanding Language Model Scaling for Protein Fitness Prediction.*
Hou C*, Di L, Zafar A, Shen Y*. *Nature Computational Science*, Jul 2026. [BioRxiv](#), April 2025.
- Protein Language Models Trained on Biophysical Dynamics Informs Mutation Effects.*
Hou C, Zhao H, Shen Y*. *PNAS*, Jan 2026. [BioRxiv](#), Oct 2024.
- PhaSepDB in 2022: Annotating Phase Separation-related Proteins with Droplet States, Co-phase Separation Partners, and Other Experimental Information.*
Hou C#, Wang X#, Xie H#, Chen T, Zhu P, Xu X, You K, Li T*. *Nucleic Acids Research*, Jan 2023.
- Systematic Prediction of Degrons and E3 Ubiquitin Ligase Binding via Deep Learning.*
Hou C, Li Y, Wang M, Wu H, Li T*. *BMC Biology*, Jul 2022.
- Screening Membraneless Organelle Participants with Machine-learning Models that Integrate Multimodal Features.*
Chen Z#, **Hou C**#, Wang L#, Yu C, Chen T, Shen B, Hou Y, Li P*, Li T*. *PNAS*, Jun 2022.
- MloDisDB: a Manually Curated Database of the Relations Between Membraneless Organelles and Diseases.*
Hou C#, Xie H#, Fu Y#, Ma Y#, Li T*. *Briefings in Bioinformatics*, Jul 2021.

Others:

9. *Expanding the Phenotypic Spectrum of MECOM-associated Syndrome: Rare Variants are Associated with Syndromic Pulmonary Arterial Hypertension.*
Welch et al. *Journal of Medical Genetics*, 2026.
10. *PhaSeDis: A Manually Curated Database of Phase Separation–Disease Associations and Corresponding Small Molecules.*
Chen et al. *Genomics Proteomics Bioinformatics*, 2025.
11. *Investigating Variant Impact on Phosphorylation Events Driving Carcinogenesis.*
Xu et al. *Briefings in Bioinformatics*, 2023.
12. *Investigating Phase Separation Properties of Chromatin-associated Proteins Using Gradient Elution of 1,6-hexanediol.*
Zhu P#, Hou C#, Liu M, Chen T, Li T*, Wang L*. *BMC Genomics*, 2023.
13. *Serum Antigenome Profiling Reveals Diagnostic Models for Rheumatoid Arthritis.*
Han P, Hou C, Zheng X, Cao L, Shi X, Zhang X, Ye H, Pan H, Liu L, Li T*, Hu F*, Li Z*. *Frontiers in Immunology*, 2022.
14. *Distinctive Network Topology of Phase-Separated Proteins in Human Interactome.*
Yu et al. *Journal of Molecular Biology*, 2022.
15. *Quantifying the Phase Separation Property of Chromatin-associated Proteins under Physiological Conditions Using an Anti-1,6-hexanediol Index.*
Shi et al. *Genome Biology*, 2021.

PRESENTATIONS

Invited & Oral Talks:

- 2026.06** *Understanding Language Model Scaling for Protein Fitness Prediction.* The University of Texas Medical Branch at Galveston.
- 2026.03** *Evolutionary and Biophysical Modeling of Protein Fitness Landscape with Language Model.* Saeed Tavazoie Lab, Columbia University, NY.
- 2026.02** *Learning Biophysical Dynamics with Language Models for Protein Function and Fitness Prediction.* Department of Systems Biology, Columbia University, NY.
- 2025.10** *Understanding Language Model Scaling on Protein Fitness Prediction.* Retreat, Department of Systems Biology, Columbia University, NY.
- 2024.09** *Learning Representation of Protein Dynamic Properties with a Language Model.* Retreat, Department of Systems Biology, Columbia University.
- 2023.07** *The Degradation Regulation of Phase Separating Proteins.* Fudan International Summer School of Life Science, Shanghai.
- 2023.06** *The Degradation Regulation of Phase Separating Proteins.* School of Basic Medical Sciences, Peking University, Beijing.
- 2023.04** *The Degradation Regulation of Phase Separating Proteins.* School of Basic Medical Sciences, Peking University, Beijing.
- 2022.12** *Targeting Disordered Degrons on Phase Separating Proteins.* Xi'an Jiaotong University.

Poster Presentations:

- 2026.05** *Understanding Language Model Scaling on Protein Fitness Prediction.* 90th Cold Spring Harbor Laboratory Symposium on Quantitative Biology: AI in Biology. Cold Spring Harbor, US.
- 2025.02** *Learning Biophysical Dynamics with Protein Language Models.* BPS 2025, Los Angeles, US.
- 2024.11** *Predicting Missense Mutation Effects with Biophysics-informed Protein Language Model.* ASHG 2024, Denver, US.
- 2024.07** *Learning Representation of Protein Dynamic Properties with a Language Model.* ISMB 2024, Montreal, Canada.

HONORS & AWARDS

- 2023** First Prize, Trainee Symposium, Fudan International Summer School
2023 Peking University Outstanding Graduate
2022 Peking University President Scholarship
2021 Peking University Doctoral Student Innovation Scholarship
2020 Peking University Junior Scholarship
2016 National Scholarship
2014 Silver Medal, National Physics Olympiad for High School Students

TEACHING & MENTORING

Rotation Ph.D. Students: Di Liu (Columbia DBMI), Aziz Zafar (Columbia DBMI)

Undergraduate Students: Elaine Cui (Columbia CS), Jason Xie (Columbia CS), Tianhao Li (PKU), Xinyi Li (PKU), Haotai Xie (PKU)

High School Students: Kevin Liu (85th Regeneron Science Talent Search Top 300 Scholar), Ludovic Pham

PROFESSIONAL SERVICE

Reviewer: EBioMedicine, PLOS Computational Biology, Protein Science, ISMB, Bioinformatics, Computational and Structural Biotechnology Journal.